

REPLICATION PACKAGE

Title of the paper: Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization

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The documents included in the Replication Package are described below.

First, the software programs used are presented. Second, the documents that should be run to generate each figure/table are described. Finally, the description of all documents is provided. Note that files are organized in different folders.

1. Software

The following software programs have been used:

- GAMS 46.5.0.
- Gurobi 11.0.0.
- MATLAB R2025a.
- Microsoft 365: Excel, PowerPoint, and Word.
- Python 3.8.8 executed in Jupyter Notebook 6.3.0 with packages NumPy 1.20.1, pandas 1.2.4, Matplotlib 3.3.4.

2. Instructions to generate the figures and tables

The instructions to generate the figures and tables are the following:

- **Figure 1:** Open figure01.pptx (root folder): PowerPoint file associated with the summary of the objective function transformations.
- **Figure 2:** Open figure02.pptx (root folder): PowerPoint file associated with the comparison of the modeling aspects of reference [13] and the proposed work.
- **Figure 3:** Open figure03.pptx (root folder): PowerPoint file associated with the decision-making tree of the proposed problem.
- **Figure 4:** Run plotFig04.m (folder “01 Scenario generation”). Document figure04.fig is generated.
- **Figure 5:** Run plotFig05.m (folder “02 Base case”). Document figure05.fig is generated.

- **Figure 6:** Run plotFig06.m (folder “02 Base case”). Document figure06.fig is generated.
- **Figure 7:** Run plotFig07.m (folder “02 Base case”). Document figure07.fig is generated.
- **Figure 8:** Run plotFig08.m (folder “02 Base case”). Document figure08.fig is generated.
- **Figure 9:** Run plotFig09.m (folder “02 Base case”). Document figure09.fig is generated.
- **Figure 10:** Run plotFig10.m (folder “02 Base case”). Document figure10.fig is generated.
- **Figure 11:** Run plotFig11.m (folder “02 Base case”). Document figure11.fig is generated.
- **Figure 12:** In Jupyter Notebook open Fig_12_Eneeco.ipynb (folder “02 Base case”) and select Run All Cells. The notebook reads the supplied input file resultsHybridSPRO_Gamma.xlsx, located in the same folder, and automatically generates and saves Figure 12 as fig_12.pdf in that folder.
- **Figure 13:** In Jupyter Notebook open Fig_13_Eneeco.ipynb (folder “02 Base case”) and select Run All Cells. The notebook reads the supplied input file resultsProposed_alpha.xlsx, located in the same folder, and automatically generates and saves Figure 13 as fig_13.pdf in that folder.
- **Figure 14:** In Jupyter Notebook open Fig_14_Eneeco.ipynb (folder “02 Base case”) and select Run All Cells. The notebook reads the supplied input file resultsHybridSPRO_Gamma.xlsx, located in the same folder, and automatically generates and saves Figure 14 as fig_14.pdf in that folder.
- **Figure 15:** In Jupyter Notebook open Fig_15_Eneeco.ipynb (folder “02 Base case”) and select Run All Cells. The notebook reads the supplied input file resultsProposed_alpha.xlsx, located in the same folder, and automatically generates and saves Figure 15 as fig_15.pdf in that folder.
- **Figure 16:** Run plotFig16.m (folder “02 Base case”). Document figure16.fig is generated.
- **Figure 17:** Run plotFig17.m (folder “02 Base case”). Document figure17.fig is generated.
- **Figure 18:** Run plotFig18.m (folder “02 Base case”). Document figure18.fig is generated.
- **Figure 19:** Run plotFig19.m (folder “02 Base case”). Document figure19.fig is generated.
- **Figure 20:** Run plotFig20.m (folder “02 Base case”). Document figure20.fig is generated.
- **Figure 21:** Run plotFig21.m (folder “02 Base case”). Document figure21.fig is generated.

- **Figure 22:** Run plotFig22.m (folder “02 Base case”). Document figure22.fig is generated.
- **Figure 23:** Run plotFig23.m (folder “02 Base case”). Document figure23.fig is generated.
- **Figure 24:** Run plotFig24.m (folder “02 Base case”). Document figure24.fig is generated.
- **Figure 25:** Run plotFig25.m (folder “02 Base case”). Document figure25.fig is generated.
- **Figure 26:** Run plotFig26.m (folder “02 Base case”). Document figure26.fig is generated.
- **Figure 27:** Run plotFig27.m (folder “06 Sensitivity analysis - MATLAB”). Document figure27.fig is generated.
- **Table 1:** Run generateTable01.m (folder “02 Base case”). Document table01.xlsx is generated.
- **Table 2:** Run generateTable02.m (folder “02 Base case”). Document table02.xlsx is generated.
- **Table 3:** Run generateTable03.m (folder “02 Base case”). Document table03.xlsx is generated.

3. Description of the documents

3.0. Root folder

- **figure01.pptx:** PowerPoint file associated with Figure 1.
- **figure02.pptx:** PowerPoint file associated with Figure 2.
- **figure03.pptx:** PowerPoint file associated with Figure 3.

3.1. Folder “01 Scenario generation”

- **data_imbalance.gms:** GAMS file that contains the realization of the imbalance price deviation. Document generated by running imbalanceScen_e2g_hour.m.
- **data_set2.gms:** GAMS file that contains the actual available production of RES used in the in-sample simulations. Document generated by running set2_mmda.m.
- **data_set3.gms:** GAMS file that contains the actual available production of RES used in the out-of-sample simulations. Document generated by running set3_pvScenarios.m.
- **figure04.fig:** MATLAB figure associated with Figure 4 generated by running plotFig04.m.

- **ImbalanceCroatia2024.xlsx**: Historical data of the imbalance price deviations.
- **imbalanceScen_e2g_hour.m**: MATLAB file that contains the code to generate the out-of-sample scenarios of the imbalance price deviations.
- **plotFig04.m**: MATLAB file used to generate Figure 4.
- **results_imbalanceScenarios.mat**: MATLAB file that contains the realization of the imbalance price deviation generated by running imbalanceScen_e2g_hour.m.
- **results_set1_pvScenarios.mat**: MATLAB file that contains the 50 PV scenarios generated by running set1_pvScenarios.m.
- **results_set2_pvScenarios.mat**: MATLAB file that contains the in-sample PV scenarios generated by running set2_mmda.m.
- **results_set3_pvScenarios.mat**: MATLAB file that contains the out-of-sample PV scenarios generated by running set3_pvScenarios.m.
- **set1_pvScenarios.m**: MATLAB file that contains the code to generate 50 PV scenarios.
- **set2_mmda.m**: MATLAB file that contains the code of the modified maximum dissimilarity algorithm to generate the in-sample PV scenarios.
- **set3_pvScenarios.m**: MATLAB file that contains the code to generate the out-of-sample PV realizations.

3.2. Folder “02 Base case”

- **codigoPausa.gms**: GAMS file used to create pauses between the generation of the results files to prevent issues.
- **cvarProfit.mat**: MATLAB file with the results of the base case generated by running plotFig17.m.
- **data.gms**: GAMS file that contains data (sets, parameters and tables).
- **data_imbalance.gms**: GAMS file that contains the realization of the imbalance price deviation.
- **data_set2.gms**: GAMS file that contains the actual available production of RES used in the in-sample simulations.
- **data_set3.gms**: GAMS file that contains the actual available production of RES used in the out-of-sample simulations.
- **dIB.mat**: MATLAB file with results exported from resultsProposed_alpha.gdx (PAR_dIB).
- **equations.gms**: GAMS file that contains the equations of the optimization models.

- **exportResultsG2E_hybrid.m**: MATLAB file used to transfer results of the previous model from GDX to XLSX to analyze them and create Figures 12 and 14.
- **exportResultsG2E_proposed.m**: MATLAB file used to transfer results of the proposed model from GDX to XLSX to analyze them and create Figures 13 and 15.
- **Fig_12_Eneeco.ipynb**: Python file used to generate Figure 12.
- **Fig_13_Eneeco.ipynb**: Python file used to generate Figure 13.
- **Fig_14_Eneeco.ipynb**: Python file used to generate Figure 14.
- **Fig_15_Eneeco.ipynb**: Python file used to generate Figure 15.
- **figure05<.fig**: MATLAB figure associated with Figure 5 generated by running plotFig05.m.
- **figure06.fig**: MATLAB figure associated with Figure 6 generated by running plotFig06.m.
- **figure07.fig**: MATLAB figure associated with Figure 6 generated by running plotFig07.m.
- **figure08.fig**: MATLAB figure associated with Figure 8 generated by running plotFig08.m.
- **figure09.fig**: MATLAB figure associated with Figure 9 generated by running plotFig09.m.
- **figure10.fig**: MATLAB figure associated with Figure 10 generated by running plotFig10.m.
- **figure11.fig**: MATLAB figure associated with Figure 11 generated by running plotFig11.m.
- **figure16.fig**: MATLAB figure associated with Figure 16 generated by running plotFig16.m.
- **figure17.fig**: MATLAB figure associated with Figure 17 generated by running plotFig17.m.
- **figure18.fig**: MATLAB figure associated with Figure 18 generated by running plotFig18.m.
- **figure19.fig**: MATLAB figure associated with Figure 19 generated by running plotFig19.m.
- **figure20.fig**: MATLAB figure associated with Figure 20 generated by running plotFig20.m.
- **figure21.fig**: MATLAB figure associated with Figure 21 generated by running plotFig21.m.
- **figure22.fig**: MATLAB figure associated with Figure 22 generated by running plotFig22.m.
- **figure23.fig**: MATLAB figure associated with Figure 23 generated by running plotFig23.m.

- **figure24.fig**: MATLAB figure associated with Figure 24 generated by running `plotFig24.m`.
- **figure25.fig**: MATLAB figure associated with Figure 25 generated by running `plotFig25.m`.
- **figure26.fig**: MATLAB figure associated with Figure 26 generated by running `plotFig26.m`.
- **generateTable01.m**: MATLAB file used to generate Table 1, which is printed in `table01.xlsx`.
- **generateTable02.m**: MATLAB file used to generate Table 2, which is printed in `table02.xlsx`.
- **generateTable03.m**: MATLAB file used to generate Table 3, which is printed in `table03.xlsx`.
- **gurobi.opt**: GAMS file that contains the options set of solver Gurobi.
- **legend_alpha.pdf**: PDF file with information about document `resultsProposed_alpha.xlsx`.
- **legend_Gamma.pdf**: PDF file with information about document `resultsHybridSPRO_Gamma.xlsx`.
- **out-of-sample-analysis-alpha.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results of the out-of-sample analysis related to Section “3.3.2. Comparison of in-sample and out-of-sample results” of the paper.
- **out-of-sample-analysis-alpha_doble_0.5.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results of the out-of-sample analysis related to a confidence level equal to 0.5 in Section “3.3.3. Impact of the confidence level” of the paper.
- **out-of-sample-analysis-alpha_doble_0.999.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results of the out-of-sample analysis related to a confidence level equal to 0.999 in Section “3.3.3. Impact of the confidence level” of the paper.
- **out-of-sample-analysis-alpha_doble_0.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results of the out-of-sample analysis related to a confidence level equal to 0 in Section “3.3.3. Impact of the confidence level” of the paper.
- **out-of-sample-analysis-alpha_S2_X3.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results of the out-of-sample analysis related to Section “3.3.1. Worst-case IPDs realization” of the paper.
- **out-of-sample-analysis-gamma_hybrid.gms**: GAMS file that contains the code to solve the previous version of the optimization problem and save the results of the out-of-sample analysis related to Section “3.3.2. Comparison of in-sample and out-of-sample results” of the paper.

- **plotFig05.m**: MATLAB file used to generate Figure 5.
- **plotFig06.m**: MATLAB file used to generate Figure 6.
- **plotFig07.m**: MATLAB file used to generate Figure 7.
- **plotFig08.m**: MATLAB file used to generate Figure 8.
- **plotFig09.m**: MATLAB file used to generate Figure 9.
- **plotFig10.m**: MATLAB file used to generate Figure 10.
- **plotFig11.m**: MATLAB file used to generate Figure 11.
- **plotFig16.m**: MATLAB file used to generate Figure 16.
- **plotFig17.m**: MATLAB file used to generate Figure 17.
- **plotFig18.m**: MATLAB file used to generate Figure 18.
- **plotFig19.m**: MATLAB file used to generate Figure 19.
- **plotFig20.m**: MATLAB file used to generate Figure 20.
- **plotFig21.m**: MATLAB file used to generate Figure 21.
- **plotFig22.m**: MATLAB file used to generate Figure 22.
- **plotFig23.m**: MATLAB file used to generate Figure 23.
- **plotFig24.m**: MATLAB file used to generate Figure 24.
- **plotFig25.m**: MATLAB file used to generate Figure 25.
- **plotFig26.m**: MATLAB file used to generate Figure 26.
- **ProfitScen.mat**: MATLAB file with results exported from results_OA_hybrid.gdx (ProfitScenHybrid).
- **results_OA.gdx**: GAMS file associated with the results generated by running out-of-sample-analysis-alpha.gms.
- **results_OA_doble_0.5.gdx**: GAMS file associated with the results generated by running out-of-sample-analysis-alpha_doble_0.5.gms.
- **results_OA_doble_0.5_export.xlsx**: Excel file with the results exported from results_OA_doble_0.5.gdx.
- **results_OA_doble_0.999.gdx**: GAMS file associated with the results generated by running out-of-sample-analysis-alpha_doble_0.999.gms.
- **results_OA_doble_0.999_export.xlsx**:
- **results_OA_doble_0.gdx**: GAMS file associated with the results generated by running out-of-sample-analysis-alpha_doble_0.gms.
- **results_OA_doble_0_export.xlsx**: Excel file with the results exported from results_OA_doble_0.gdx.
- **results_OA_export.xlsx**: Excel file with the results exported from results_OA.gdx.
- **results_OA_hybrid.gdx**: GAMS file associated with the results generated by running out-of-sample-analysis-gamma_hybrid.gms.
- **results_OA_S2_X3.gdx**: GAMS file associated with the results generated by running out-of-sample-analysis-alpha_S2_X3.gms.
- **results_OA_S2_X3_export.xlsx**:

- **results_set2_pvScenarios.mat**: MATLAB file used to load data in plotFig16.m.
- **resultsHybridSPRO_Gamma.gdx**: GAMS file associated with the results generated by running solveHybridSPRO_Gamma.gms.
- **resultsHybridSPRO_Gamma.xlsx**: Excel file with the results of Figures 12 and 14.
- **resultsHybridSPRO_Gamma_export.xlsx**:
- **resultsProposed_alpha.gdx**: GAMS file associated with the results generated by running solveProposed_alpha.gms.
- **resultsProposed_alpha.xlsx**: Excel file with the results of Figures 13 and 15.
- **resultsProposed_alpha_export.xlsx**: Excel file with the results exported from resultsProposed_alpha.gdx.
- **solveHybridSPRO.gms**: GAMS file that contains the code to solve the previous version of the optimization problem and save the results related to Section “3.2. In-sample analyses” of the paper.
- **solveHybridSPRO_Gamma.gms**: GAMS file that contains the code to solve the previous version of the optimization problem and save the results for different values of the uncertainty budget related to Section “3.2. In-sample analyses” of the paper.
- **solveProposed.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results related to Section “3.2. In-sample analyses” of the paper.
- **solveProposed_alpha.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results for different values of the confidence level related to Section “3.2. In-sample analyses” of the paper.
- **table01.xlsx**: Excel file with the results linked to Table 1 generated by running generateTable01.m.
- **table02.xlsx**: Excel file with the results linked to Table 2 generated by running generateTable02.m.
- **table03.xlsx**: Excel file with the results linked to Table 3 generated by running generateTable03.m.
- **variables.gms**: GAMS file that contains the variables of the optimization models.
- **y.mat**: MATLAB file with results exported from resultsProposed_alpha.gdx (PAR_y).
- **zXI.mat**: MATLAB file with results exported from resultsProposed_alpha.gdx (PAR_zXI).

3.3. Folder “03 Sensitivity analysis 1”

- **codigoPausa.gms**: GAMS file used to create pauses between the generation of the results files to prevent issues.
- **cvarProfit_1.mat**: MATLAB file with the results of the first sensitivity analysis generated by running obtainCVaR.m.
- **data.gms**: GAMS file that contains data (sets, parameters and tables).
- **data_imbalance.gms**: GAMS file that contains the realization of the imbalance price deviation.
- **data_set2.gms**: GAMS file that contains the actual available production of RES used in the in-sample simulations.
- **data_set3.gms**: GAMS file that contains the actual available production of RES used in the out-of-sample simulations.
- **equations.gms**: GAMS file that contains the equations of the optimization models.
- **gurobi.opt**: GAMS file that contains the options set of solver GUROBI.
- **obtainCVaR.m**: MATLAB file used to generate cvarProfit_1.mat, which is later used to generate Figure 27 in folder “06 Sensitivity analysis – MATLAB”.
- **out-of-sample-analysis-alpha.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results of the out-of-sample analysis related to Section “3.3.2. Comparison of in-sample and out-of-sample results” of the paper.
- **results_OA.gdx**: GAMS file associated with the results generated by running out-of-sample-analysis-alpha.gms.
- **results_OA_export.xlsx**: Excel file with the results exported from results_OA.gdx.
- **solveProposed.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results.
- **solveProposed_alpha.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results for different values of the confidence level.
- **variables.gms**: GAMS file that contains the variables of the optimization models.

3.4. Folder “03 Sensitivity analysis 2”

- **codigoPausa.gms**: GAMS file used to create pauses between the generation of the results files to prevent issues.
- **cvarProfit_2.mat**: MATLAB file with the results of the second sensitivity analysis generated by running obtainCVaR.m.
- **data.gms**: GAMS file that contains data (sets, parameters and tables).

- **data_imbalance.gms**: GAMS file that contains the realization of the imbalance price deviation.
- **data_set2.gms**: GAMS file that contains the actual available production of RES used in the in-sample simulations.
- **data_set3.gms**: GAMS file that contains the actual available production of RES used in the out-of-sample simulations.
- **equations.gms**: GAMS file that contains the equations of the optimization models.
- **gurobi.opt**: GAMS file that contains the options set of solver GUROBI.
- **obtainCVaR.m**: MATLAB file used to generate cvarProfit_2.mat, which is later used to generate Figure 27 in folder “06 Sensitivity analysis – MATLAB”.
- **out-of-sample-analysis-alpha.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results of the out-of-sample analysis related to Section “3.3.2. Comparison of in-sample and out-of-sample results” of the paper.
- **results_OA.gdx**: GAMS file associated with the results generated by running out-of-sample-analysis-alpha.gms.
- **results_OA_export.xlsx**: Excel file with the results exported from results_OA.gdx.
- **solveProposed.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results.
- **solveProposed_alpha.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results for different values of the confidence level.
- **variables.gms**: GAMS file that contains the variables of the optimization models.

3.5. Folder “03 Sensitivity analysis 3”

- **codigoPausa.gms**: GAMS file used to create pauses between the generation of the results files to prevent issues.
- **cvarProfit_3.mat**: MATLAB file with the results of the third sensitivity analysis generated by running obtainCVaR.m.
- **data.gms**: GAMS file that contains data (sets, parameters and tables).
- **data_imbalance.gms**: GAMS file that contains the realization of the imbalance price deviation.
- **data_set2.gms**: GAMS file that contains the actual available production of RES used in the in-sample simulations.
- **data_set3.gms**: GAMS file that contains the actual available production of RES used in the out-of-sample simulations.

- **equations.gms**: GAMS file that contains the equations of the optimization models.
- **gurobi.opt**: GAMS file that contains the options set of solver GUROBI.
- **obtainCVaR.m**: MATLAB file used to generate cvarProfit_3.mat, which is later used to generate Figure 27 in folder “06 Sensitivity analysis – MATLAB”.
- **out-of-sample-analysis-alpha.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results of the out-of-sample analysis related to Section “3.3.2. Comparison of in-sample and out-of-sample results” of the paper.
- **results_OA.gdx**: GAMS file associated with the results generated by running out-of-sample-analysis-alpha.gms.
- **results_OA_export.xlsx**: Excel file with the results exported from results_OA.gdx.
- **solveProposed.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results.
- **solveProposed_alpha.gms**: GAMS file that contains the code to solve the proposed optimization problem and save the results for different values of the confidence level.
- **variables.gms**: GAMS file that contains the variables of the optimization models.

3.6. Folder “06 Sensitivity analysis - MATLAB”

- **cvarProfit.mat**: MATLAB file with the results of the base case generated by running plotFig17.m (folder “02 Base case”).
- **cvarProfit_1.mat**: MATLAB file with the results of the first sensitivity analysis generated by running obtainCVaR.m (folder “03 Sensitivity analysis 1”).
- **cvarProfit_2.mat**: MATLAB file with the results of the second sensitivity analysis generated by running obtainCVaR.m (folder “04 Sensitivity analysis 2”).
- **cvarProfit_3.mat**: MATLAB file with the results of the third sensitivity analysis generated by running obtainCVaR.m (folder “05 Sensitivity analysis 3”).
- **figure27.fig**: MATLAB figure associated with Figure 27 generated by running plotFig27.m.
- **plotFig27.m**: MATLAB file used to generate Figure 27.
- **resultsProposed_alpha_1.gdx**: GAMS file associated with the results generated by running solveProposed_alpha.gms (folder “03 Sensitivity analysis 1”).
- **resultsProposed_alpha_1_export.xlsx**: Excel file with the results exported from resultsProposed_alpha_1.gdx.

- **resultsProposed_alpha_2.gdx**: GAMS file associated with the results generated by running solveProposed_alpha.gms (folder “04 Sensitivity analysis 2”).
- **resultsProposed_alpha_2_export.xlsx**: Excel file with the results exported from resultsProposed_alpha_2.gdx.
- **resultsProposed_alpha_3.gdx**: GAMS file associated with the results generated by running solveProposed_alpha.gms (folder “05 Sensitivity analysis 3”).
- **resultsProposed_alpha_3_export.xlsx**: Excel file with the results exported from resultsProposed_alpha_3.gdx.